

| Assessment | Test | Domain                            | Skill                                                | Difficulty                                        |
|------------|------|-----------------------------------|------------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Ratios, rates, proportional relationships, and units | Easy <div><div></div><div></div><div></div></div> |

Question

A cable provider wanted to know how many of its **30,000** customers would be interested in a new service plan. The provider selected **300** customers at random and asked each customer whether the customer would be interested in the new plan. Of those surveyed, **8** said they would be interested. Which of the following is the best estimate of the total number of customers who would be interested in the new service plan?

Answer

- A. 8
- B. 80
- C. 800
- D. 8,000

Correct Answer: C

Rationale

Choice C is correct. It’s given that **8** out of **300** customers in a random sample said they would be interested in the new plan. This means the fraction of sampled customers who said they would be interested is  $\frac{8}{300}$ . Multiplying this fraction by the total customer population yields  $(\frac{8}{300})(30,000)$ , or **800**. Therefore, the best estimate of the total number of customers who would be interested in the new service plan is **800**.

Choice A is incorrect. This is the number of customers surveyed who said they would be interested in the new plan, not the best estimate of the total number of customers who would be interested in the new plan.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.